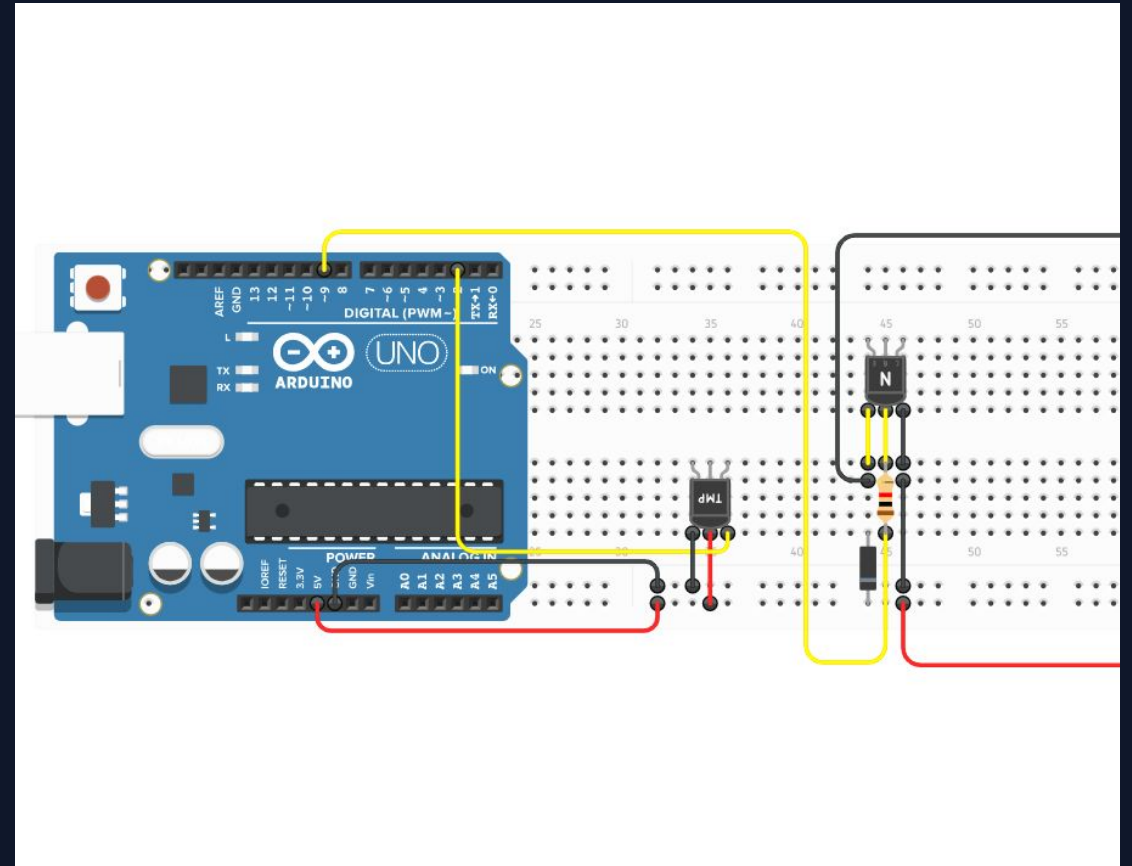


Fan System

Arduino + DHT11 + PWM motor control

Goal: automatically increase fan speed when the temperature rises.

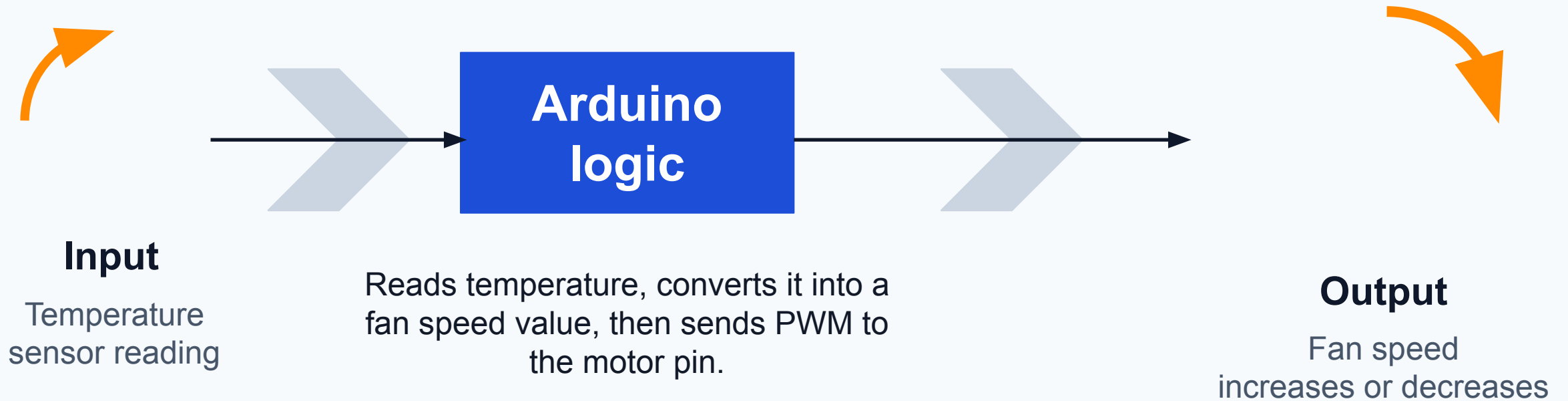
Tsukasa Takahashi, Zihan Chen, Jeremiah Dacanay, Mason Curcuru



When the sensor reads a higher temperature, the Arduino sends a stronger PWM signal to the motor driver so the fan spins faster.

Project Concept

What the system is designed to do

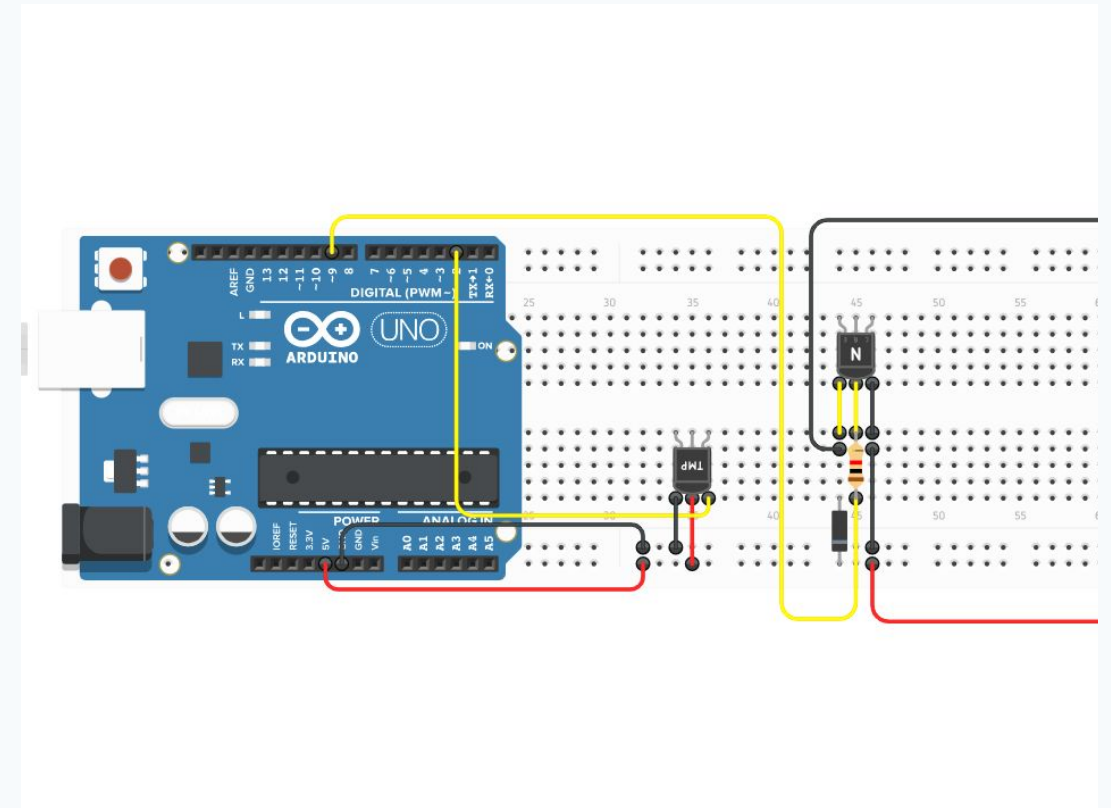


Simple control idea: hotter air = faster fan.

Main Parts Used

Each part has a specific job in the circuit

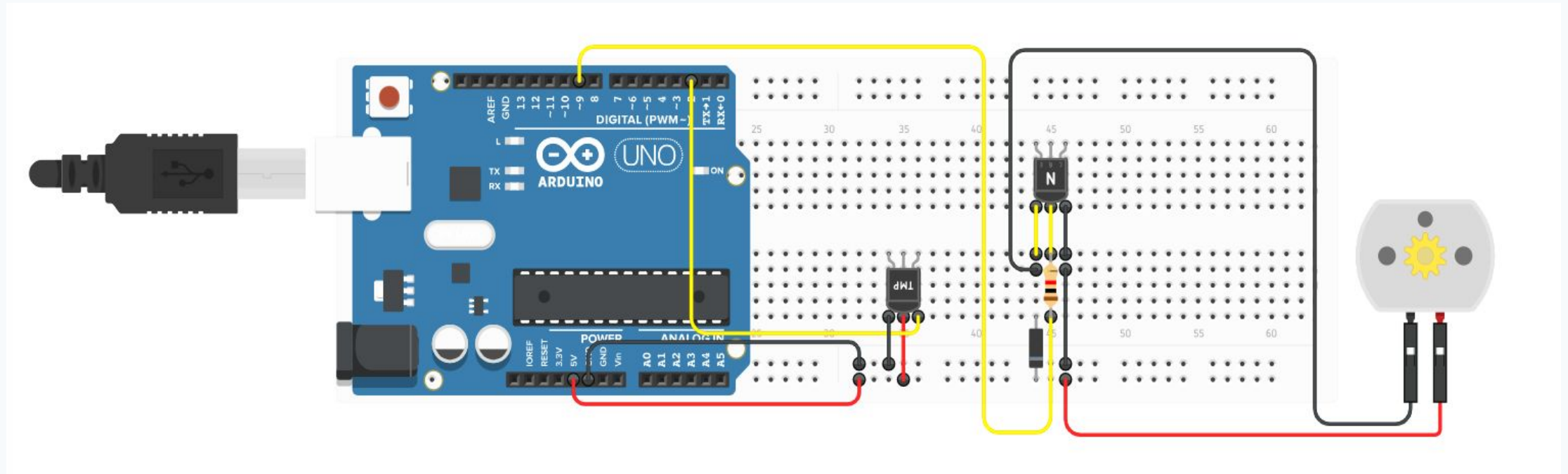
- 1 Arduino Uno: runs the code and outputs the PWM control signal.
- 2 DHT11 temperature sensor: measures the surrounding air temperature.
- 3 Transistor / MOSFET driver: lets the Arduino safely control the motor current.
- 4 DC motor with fan: converts the electrical signal into airflow.
- 5 Diode and resistor: protect/control the motor driver circuit.



Circuit overview

Circuit Wiring

How the sensor and fan connect to the Arduino



Power and ground complete the circuit

DHT11 sensor reads temperature

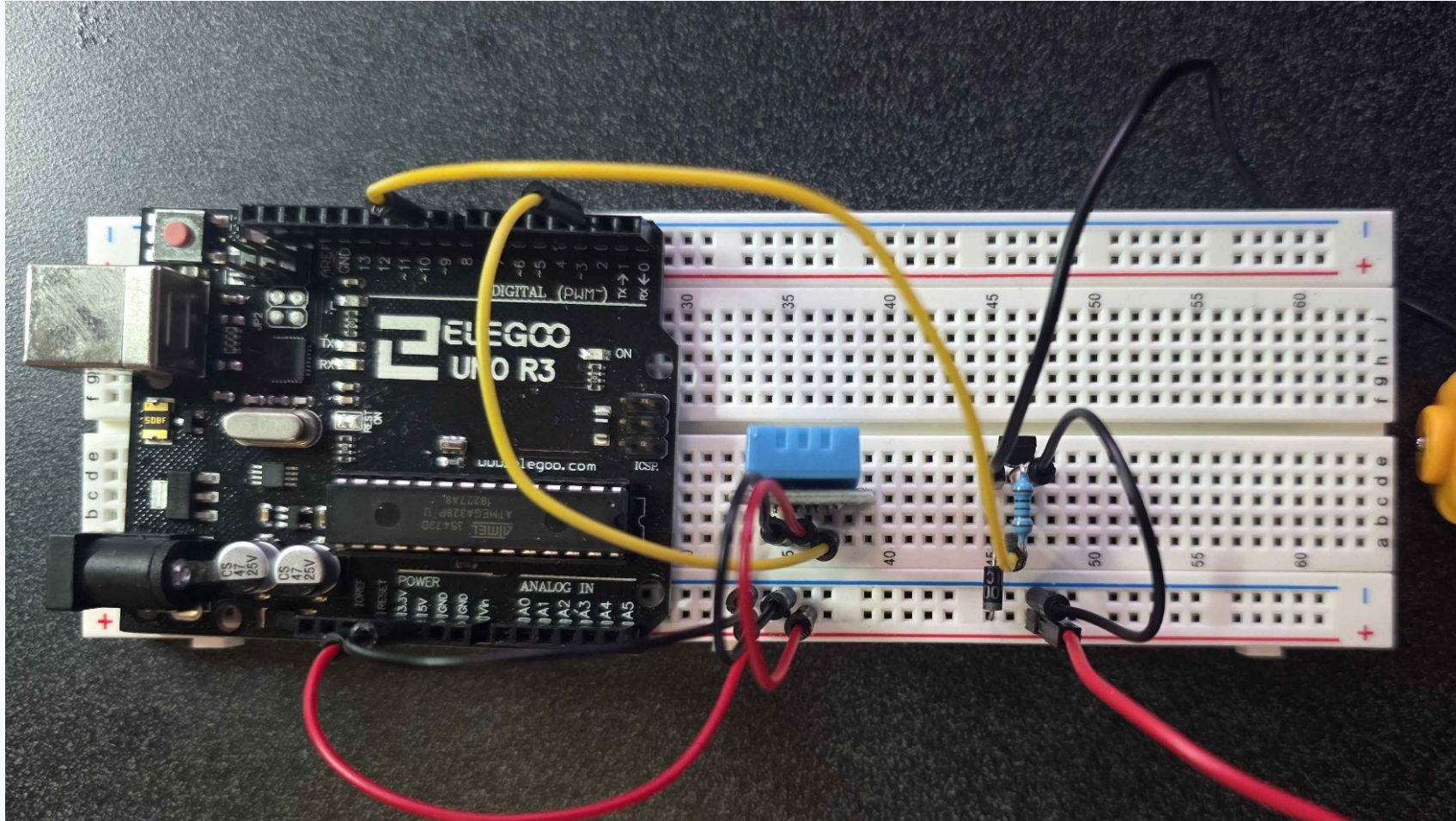
Pin 9 sends PWM to fan driver

The motor is not driven directly from the Arduino pin. The control pin tells the transistor/MOSFET how much power to send to the fan.

⌘ The short paths to PN2222 are shown this way for visual clarity; they are actually compact

Circuit Wiring

actual circuit



Build Process

From assignment idea to working prototype

1. Temperature-Controlled Fan System

Sensors: LM35 or DHT22

Concepts Covered:

- Analog signal reading
- ADC conversion
- Transistor as a switch
- PWM motor control

How it works:

- Arduino reads temperature.
- If temperature rises, fan speed increases.
- Uses transistor or MOSFET to drive motor.

Starting point



```
fan_system_code.ino
1  #include "DHT.h"
2
3  #define DHTPIN 2
4  #define DHTTYPE DHT11
5  #define MOTORPIN 9
6
7  DHT dht(DHTPIN, DHTTYPE);
8
9  void setup() {
10     Serial.begin(9600);
11     dht.begin();
12     pinMode(MOTORPIN, OUTPUT);
13 }
14
15 void loop() {
16     float temp = dht.readTemperature();
17
18     if (isnan(temp)) {
19         Serial.println("Failed to read from DHT se
20         return;
21     }
22
23     int fanSpeed = map(temp, 23, 26, 0, 255);
24     fanSpeed = constrain(fanSpeed, 0, 255);
25
26     analogWrite(MOTORPIN, fanSpeed);
27
28     Serial.print("Temperature: ");
29     Serial.print(temp);
```

Code created



Test

Warm the sensor
→ temperature rises
→ fan speed increases

The project combined the circuit build, Arduino programming, and live testing through the Serial Monitor.

Code: Setup and Sensor Reading

The Arduino first prepares the sensor and motor pin

```
#include "DHT.h"

#define DHTPIN 2
#define DHTTYPE DHT11
#define MOTORPIN 9

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
  pinMode(MOTORPIN, OUTPUT);
}
```

What this part does

Loads the DHT temperature sensor library.

Sets the temperature sensor to digital pin 2.

Sets the motor control output to PWM pin 9.

Starts Serial Monitor so we can see temperature and fan speed.

Setup happens once when the Arduino starts.

Code: Fan Speed Control

The loop reads temperature and converts it into motor speed

```
void loop() {  
  float temp = dht.readTemperature();  
  
  if (isnan(temp)) {  
    Serial.println("Failed to read");  
    return;  
  }  
  
  int fanSpeed = map(temp, 23, 26, 0, 255);  
  fanSpeed = constrain(fanSpeed, 0, 255);  
  
  analogWrite(MOTORPIN, fanSpeed);  
  delay(2000);  
}
```

Control logic



The Arduino converts temperature into a PWM number. PWM values range from 0 to 255, which controls how much power the fan receives.

Testing and Results

What we looked for during testing

```
fan_system_code.ino
1  #include "DHT.h"
2
3  #define DHTPIN 2
4  #define DHTTYPE DHT11
5  #define MOTORPIN 9
6
7  DHT dht(DHTPIN, DHTTYPE);
8
9  void setup() {
10     Serial.begin(9600);
11     dht.begin();
12     pinMode(MOTORPIN, OUTPUT);
13 }
14
15 void loop() {
16     float temp = dht.readTemperature();
17
18     if (isnan(temp)) {
19         Serial.println("Failed to read from DHT sensor!");
20         return;
21     }
22
23     int fanSpeed = map(temp, 23, 26, 0, 255);
24     fanSpeed = constrain(fanSpeed, 0, 255);
25
26     analogWrite(MOTORPIN, fanSpeed);
27
28     Serial.print("Temperature: ");
29     Serial.print(temp);
30     Serial.print(" C, Fan Speed: ");
31     Serial.println(fanSpeed);
32
33     delay(2000);
34 }
```

Temperature reading appears in Serial Monitor

Fan speed value changes from 0 to 255

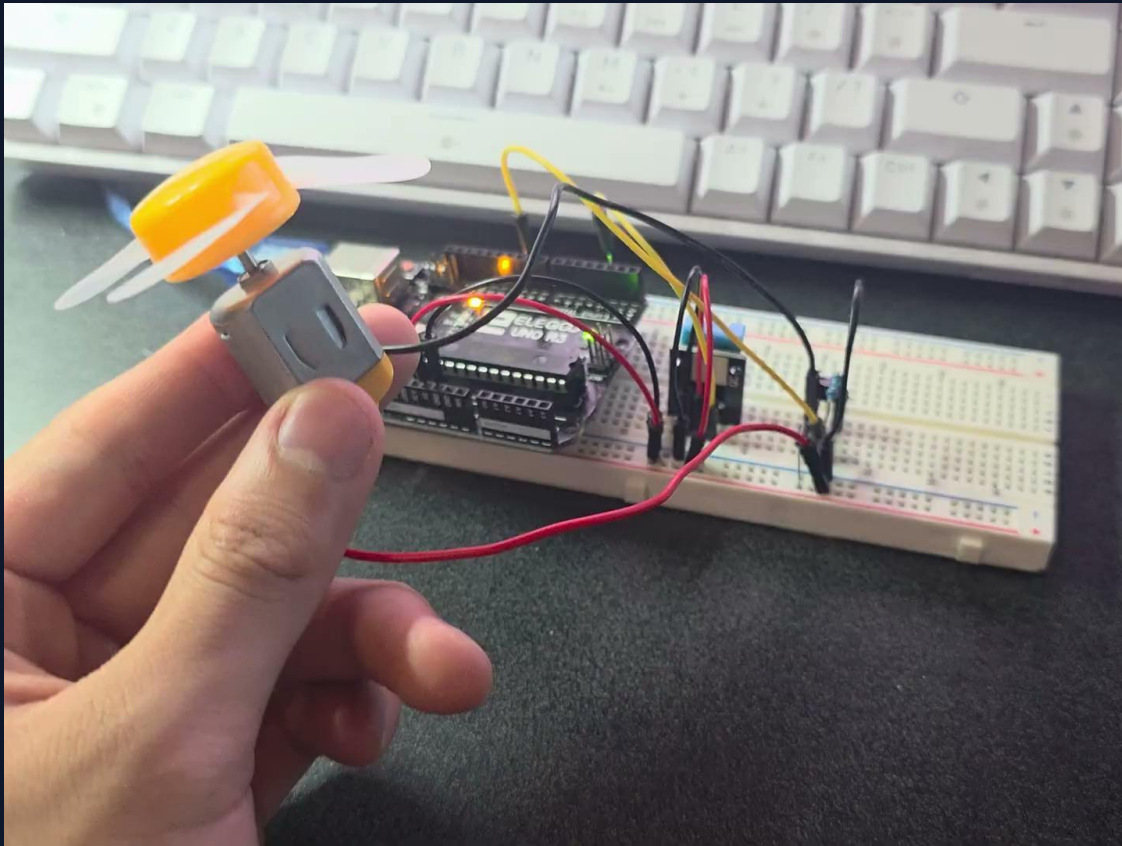
Fan should spin faster as temperature rises

This confirmed the Arduino was receiving sensor data and sending a changing PWM signal to control the motor.

Testing and Results

A scene in operation

Operating



Actual behavior

Once it started operating, the DHT11 measured room temperature and since it was between 23 and 26 degrees, set the fan to rotate a moderate speed

At around the 28-second mark in the video, hot air was blown onto the DHT11

the sensor then detected a rise in temperature and increased the fan speed at around the 31-second mark in the video

Improvements and Conclusion

What worked, and what could be improved next

Conclusion

Our temperature-controlled fan shows how a sensor, Arduino code, and motor driver can work together to create an automatic cooling system.

The fan changes speed based on the temperature reading instead of running at one constant speed.

Possible improvements

Add an LCD screen to show the temperature digitally.

Use a better temperature range or fan curve for smoother control.

Use a separate motor power supply for stronger fan performance.

Build the final circuit on a cleaner, more permanent board.